

assignment #02

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CHE 301 – Assignment 2 Report

Regenerative Rankine Cycle

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Abstract

This report documents the solution approach for CHE 301 Assignment 2 involving a regenerative Rankine cycle. Stream properties are obtained using the CHE Calculator, and each device is analyzed using steady-state material, energy, and entropy balances with negligible kinetic and potential energy changes. Numerical steps such as Goal Seek are used to determine unknown outlet states.

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1 Background

This assignment considers a regenerative Rankine cycle for power production. High-pressure steam expands through turbines, condenses in a condenser, and is pumped back to the boiler. A portion of steam is extracted and mixed in an open feed water heater to improve performance. The required task is to compute stream properties and performance metrics using macroscopic balances and the CHE Calculator.

Nomenclature / Symbol List

$\dot{n}$	Molar flow rate
h	Liquid/Vapor enthalpy
s	Liquid/Vapor entropy
H	Molar enthalpy
S	Molar entropy
x	Quality
$\dot{Q}$	Heat transfer rate
$\dot{W}$	Work / power
B	Availability
LW	Lost work
T_0	Ambient temperature

2 Assignment Description and Given Information

The assignment considers a regenerative Rankine cycle for power production. The boiler (F-100) generates steam at a pressure of 40 bar and a temperature of 450°C. The superheated steam expands in the High Pressure (HP) steam turbine (ST-100) down to 2 bar. A fraction of the turbine discharge is extracted and directed to an Open Feedwater Heater (OFWH), while the remaining flow continues to the Low Pressure (LP) turbine (ST-101).

The discharge from the LP turbine is sent to a vacuum condenser (E-100), where the stream leaves as saturated liquid at 40°C. The condensate is then pressurized using a condensate pump before mixing with the extracted steam in the OFWH. The mixed stream exits as a subcooled liquid and is pressurized again by a feedwater pump before returning to the boiler, completing the regenerative Rankine cycle.

The purpose of this assignment is to determine the thermodynamic properties of each stream, evaluate the performance of each device, and calculate efficiencies, lost work, and utility requirements using macroscopic balances and property data obtained from the CHE Calculator.

The schematic of the process flow diagram is shown below and will be used as a reference for all stream numbers and device analyses.

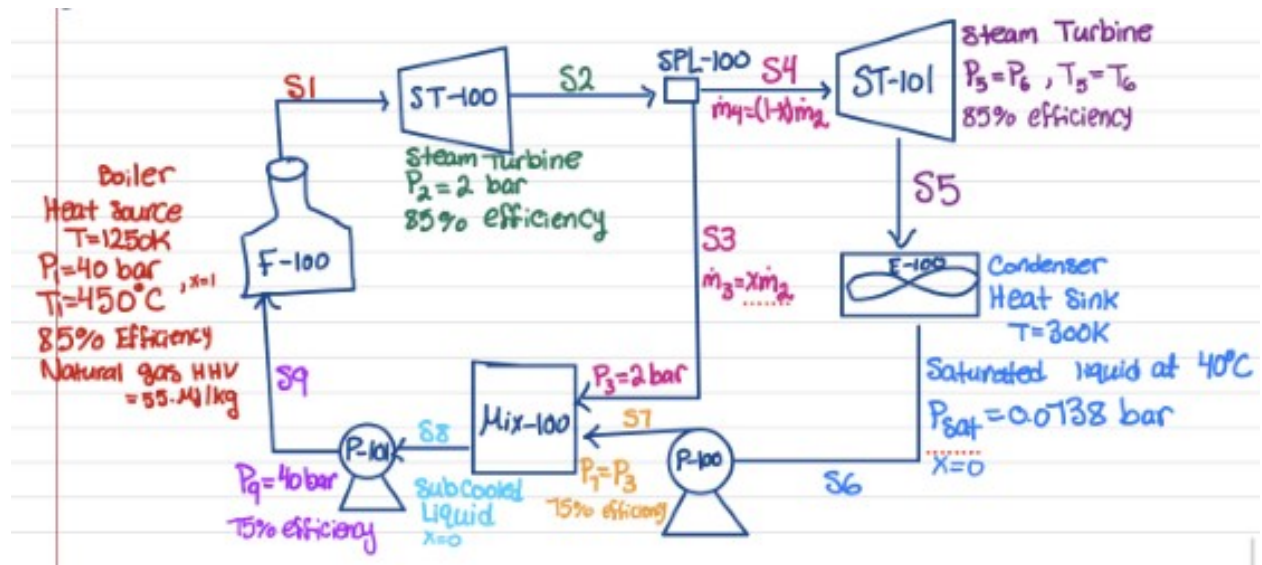


Figure 1: Hand Drawn Schematics of Full Process

3 Modeling Assumptions and Solver Strategy

3.1 Assumptions

- steady state
- $\Delta \text{PE} = \Delta \text{KE} = 0$

3.2 Macroscopic Balance Equations

Material balance:

$$\sum \dot{n}_{in} = \sum \dot{n}_{out} \quad (1)$$

Energy balance:

$$0 = \dot{Q} - \dot{W} + \sum \dot{n}_{in} H_{in} - \sum \dot{n}_{out} H_{out} \quad (2)$$

Entropy balance:

$$0 = \sum \dot{n}_{in} S_{in} - \sum \dot{n}_{out} S_{out} + \sum_j \frac{\dot{Q}_j}{T_j} + \dot{S}_{gen} \quad (3)$$

Availability:

$$B = \dot{H} - T_0 \dot{S} \quad (4)$$

3.3 Solver Strategy

Thermodynamic properties are obtained using CHE Calculator functions such as **HSteam** and **SSteam**. When unknown outlet states are required, an entropy balance is used for the isentropic reference case and an efficiency balance is used for the real device. Numerical methods such as Goal Seek are applied to drive the balance to zero by adjusting temperature or quality.

4 Device Analyses

In order to proceed with this assignment we have to find the properties for each stream in this Rankine Cycle design. To do so we will do an analysis of each device in the cycle. All the calculations in this report that are used in the analyzation of each device and in the solving of this problem were calculated using the CHE Calculator.

4.1 The Analysis of the Boiler: Finding H and S

The boiler is given at 40 bar and 450°C. We know that it is superheated vapor/steam. So to find the entropy and enthalpy we use the CHE Calculator as follows:

To find the enthalpies:

$$\text{HSteam}(T,P) \Rightarrow H_{\text{steam}}(450, 40) = 60011.902 \text{ kJ/kmol} \quad (5)$$

To find the entropies:

$$\text{SSteam}(T,P) \Rightarrow S_{\text{steam}}(450, 40) = 125.003 \text{ kJ/(kmol} \cdot \text{K)} \quad (6)$$

To find the molar enthalpy and entropy we will use the quality, x , since it is superheated vapor we know $x = 1$ so,

Molar enthalpy:

$$H = h_g x + h_f (1 - x) \quad (7)$$

$$H = (60011.902)(1) + 0 = 60011.902 \text{ kJ/kmol} \quad (8)$$

Molar entropy:

$$S = s_g x + s_f (1 - x) \quad (9)$$

$$S = (125.003)(1) + 0 = 125.003 \text{ kJ/(kmol} \cdot \text{K)} \quad (10)$$

Now to find the enthalpy and entropy flow you will need to plug in a random number for the molar flow rate, let's plug in 1 kmol/s. So now multiply the molar enthalpy and entropy by this molar flow rate and get those numbers, they will change when we find the correct molar flow rate.

Now find the availability, which is:

$$B = \dot{H} - T_0 \dot{S} \quad (11)$$

B = Enthalpy Flow – Temperature of the infinite universe $\times$ Entropy Flow where the Temperature of the infinite universe is $T_0 = 300$ K. Calculate this number, again it will change once we have the correct molar flow rate which we will calculate for after we analyze every stream.

13	Stream Properties	S1
14	Molar Flow, kmol/s	0.0289
15	Mass Flow, kg/s	
16	Volume Flow, m³/s	
17	Temperature, C	450.0000
18	Pressure, bar	40.0000
19	Vapor Fraction	1.0000
20	Molecular Weight, kg/kmol	
21		
22	Molar Enthalpy, kJ/kmol	60011.902
23	Vapor Phase	60011.902
24	Liquid Phase	
25	Molar Entropy, kJ/(kmol K)	125.003
26	Vapor Phase	125.003
27	Liquid Phase	
28	Molar Density, kmol/m³	
29		
30	Enthalpy Flow, kJ/s	1737.15
31	Entropy Flow, kJ/(s K)	3.62
32	Availability, kJ/s	651.62
33	Duty [Power], kJ/s [kW]	
34	Efficiency	
35		
36	Compressibility Factor	
37		

Figure 2: Stream 1

4.2 The Analysis of High Pressure Turbine: Finding T_2 , x , H , S

First we know that the turbine will be an isentropic Process so therefore the temperature at the imaginary outlet stream of the turbine will be the saturation temperature at 2 bar before we can find the quality x , we need to find the enthalpies and entropies of the imaginary outlet stream using the CHE Calculator. We know it will be a mixture so to find the right numbers we would need to put in the calculator as follows:

To find the enthalpies:

$$\text{HSteam(P,V)} \Rightarrow H_{\text{steam}}(2, v) = 48753.8215 \text{ kJ/kmol} \quad (12)$$

$$\text{HSteam(P,L)} \Rightarrow H_{\text{steam}}(2, l) = 9092.186 \text{ kJ/kmol} \quad (13)$$

To find the entropies:

$$S_{\text{steam}}(2, v) = 128.390 \text{ kJ}/(\text{kmol} \cdot \text{K}) \quad (14)$$

$$S_{\text{steam}}(2, l) = 27.564 \text{ kJ}/(\text{kmol} \cdot \text{K}) \quad (15)$$

Now to find the molar enthalpy and entropy you would have to plug any number for the quality x where $0 < x < 1$ and then use it to solve the equation for molar enthalpy and molar entropy:

Molar Enthalpy:

$$H = h_g x + h_f (1 - x) \quad (16)$$

Molar Entropy:

$$S = s_g x + s_f (1 - x) \quad (17)$$

Plug in 1 for x to give you a number. Now to find the right number for quality we need to make an entropy balance such as:

$$\Delta S = S_{\text{in}} - S_{\text{out}} \quad (18)$$

S_{in} is the molar entropy of Stream 1 and S_{out} is the molar entropy of stream 2.

Once you get a number for this equation you will use the numerical method goal seek to set the entropy balance to equal zero by changing the quality x of the imaginary outlet stream

of the HP Turbine. When we did that we got a quality of:

$$x = 0.9664 \quad (19)$$

Now to find the enthalpy and entropy flow and the availability calculate them the same way as we did in the boiler analysis.

We can now find the duty (power) for the isentropic stream of the turbine and we can do that by using the following formula:

$$\dot{W} = H_{\text{out}} - H_{\text{in}} \quad (20)$$

and get a number. This is an imaginary power so we won't need it for the calculations of the next streams.

Now to find the properties of the real outlet stream of the HP Turbine we will do the same process as in the imaginary stream to find the entropies and enthalpies for the real stream of the HP Turbine, plug in random numbers for the x , where $0 < x < 1$, and T_2 , and get the molar enthalpies and entropies, and now we can find the enthalpy and entropy flows, duty and availability. We can also find the efficiency using:

$$\text{Efficiency balance} = \frac{\dot{W}_{\text{actual}}}{\dot{W}_{\text{ideal}}} \quad (21)$$

We now can do an efficiency balance such as:

Efficiency balance = Calculated Efficiency- Given Efficiency

$$\text{Efficiency balance} = \frac{\dot{W}_{\text{actual}}}{\dot{W}_{\text{ideal}}} - 0.85 \quad (22)$$

Once we have a number we'll have to goal seek the quality to see what state it is in once we do that we notice that the $x > 1$ so that tells us that it is a superheated vapor so insted we just plug in 1 for the quality and goal seek the temperature so the efficiency balance can equal zero. When we do that we find that the temperature is:

$$T_2 = 134.8995^\circ\text{C} \quad (23)$$

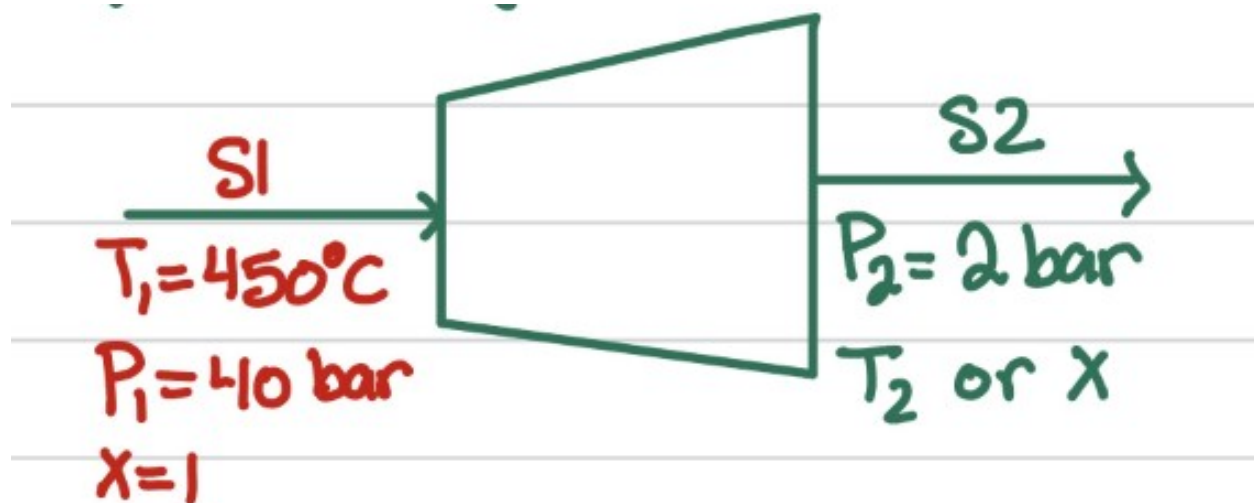


Figure 3: HP Turbine

Stream Properties	S2 Ideal	S2
Molar Flow, kmol/s	0.0289	0.0289
Mass Flow, kg/s		
Volume Flow, m ³ /s		
Temperature, C	120.2308	134.8995
Pressure, bar	2.0000	2.0000
Vapor Fraction	0.9664	1.0000
Molecular Weight, kg/kmol		
Molar Enthalpy, kJ/kmol	47421.451	49310.021
Vapor Phase	48753.8215	49310.021
Liquid Phase	9092.186	
Molar Entropy, kJ/(kmol K)	125.003	129.778
Vapor Phase	128.390	129.778
Liquid Phase	27.564	
Molar Density, kmol/m ³		
Enthalpy Flow, kJ/s	1372.70	1427.36
Entropy Flow, kJ/(s K)	3.62	3.76
Availability, kJ/s	287.17	300.37
Duty [Power], kJ/s [kW]	-364.45	-309.78
Efficiency		0.85
Compressibility Factor		

Figure 4: Stream 2: Ideal and Real

4.3 The Analysis of the Splitter

The splitter does not change the phase or the temperatures or the pressures, it just splits the product into multiple streams. So the properties of the two streams at the outlet of the splitter will be the exact same except the molar flow rate which we will discuss how we got the mass fraction for it later on in the report.

Now the properties of streams 3 and 4 are as follows:

- Superheated vapor, $x = 1$
- $T_3 = T_4 = T_2 = 134.8995^\circ\text{C}$
- $P_3 = P_4 = P_2 = 2 \text{ bar}$
- $H_3 = H_4 = H_2 = 49310.021 \text{ kJ/kmol}$
- $S_3 = S_4 = S_2 = 129.378 \text{ kJ}/(\text{kmol} \cdot \text{K})$

For the molar entropy and enthalpy and availability will be calculated just like in the calculations for the boiler and the turbine. The duty (power) is different from all the devices because the splitter doesn't use or produce work or heat so the duty will be zero. For the molar flow rate we would be relying on other streams and devices, the molar flow rates for each stream will be as follows:

$$\dot{n}_1 = \dot{n}_2 \quad (24)$$

$$\dot{n}_3 = x \dot{n}_2, \text{ where } x \text{ is fraction of what stream 2 produce} \quad (25)$$

$$\dot{n}_4 = (1 - x) \dot{n}_2 = \dot{n}_5 = \dot{n}_6 = \dot{n}_7 \quad (26)$$

$$\dot{n}_8 = \dot{n}_3 + \dot{n}_7, \text{ Since there are two inlets and one outlet} \quad (27)$$

$$\dot{n}_8 = \dot{n}_3 + \dot{n}_7 = \dot{n}_9 \quad (28)$$

We can use those flow rates and set them that way because it is assumed that this cycle is at steady state and therefore $\dot{n}_{in} = \dot{n}_{out}$ and that applies to the whole cycle.

In order to find the fraction we will need to use data from other stream in this cycle so in this section we will derive the formula and we will evaluate the answer later.

So to calculate the mass fraction we will use stream 8, the outlet of the mixture.

Mass and Energy Balance Derivation for the Fraction

Steady state so:

$$\dot{n}_{in} = \dot{n}_{out} \quad (29)$$

Energy balance:

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \sum \dot{n}_{in} H_{in} - \sum \dot{n}_{out} H_{out} \quad (30)$$

Steady state, no heat or work, and $\Delta PE = \Delta KE = 0$, so the equation will be below:

$$0 = \sum \dot{n}_{in} H_{in} - \sum \dot{n}_{out} H_{out} \quad (31)$$

Set equal to each other:

$$\dot{n}_8 H_8 = \dot{n}_7 H_7 + \dot{n}_3 H_3 \quad (32)$$

We know that $\dot{n}_3 = x\dot{n}_2$ and $\dot{n}_7 = (1 - x)\dot{n}_2$, so using the molar mass balance $\dot{n}_8 = \dot{n}_7 + \dot{n}_3$ we can substitute for $\dot{n}_8$:

$$[x\dot{n}_2 + (1 - x)\dot{n}_2] H_8 = (1 - x)\dot{n}_2 H_7 + x\dot{n}_2 H_3 \quad (33)$$

We can divide by $\dot{n}_2$ throughout:

$$H_8 = (1 - x)H_7 + xH_3 \quad (34)$$

Now we use the energy balance and substitute all those formulas in the use:

$$xH_8 + (1 - x)h_8 = (1 - x)H_7 + xH_3 \quad (35)$$

$$xH_8 - H_8 = xH_3 - xH_7 - H_7 + H_8 \quad (36)$$

$$H_8 - H_7 = x(H_3 - H_7) \quad (37)$$

$$x = \frac{H_8 - H_7}{H_3 - H_7} \quad (38)$$

Stream Properties	S3	S4
Molar Flow, kmol/s	0.0037	0.0252
Mass Flow, kg/s		
Volume Flow, m ³ /s		
Temperature, C	134.8995	134.8995
Pressure, bar	2.0000	2.0000
Vapor Fraction	1.0000	1.0000
Molecular Weight, kg/kmol		
Molar Enthalpy, kJ/kmol	49310.021	49310.021
Vapor Phase	49310.021	49310.021
Liquid Phase		
Molar Entropy, kJ/(kmol K)	129.778	129.778
Vapor Phase	129.778	129.778
Liquid Phase		
Molar Density, kmol/m ³		
Enthalpy Flow, kJ/s	184.84	1242.52
Entropy Flow, kJ/(s K)	0.49	3.27
Availability, kJ/s	38.90	261.47
Duty [Power], kJ/s [kW]	0.00	0.00
Efficiency		
Compressibility Factor		

Figure 5: Streams 3 and 4

4.4 The Analysis of the Low Pressure Turbine

We know that the turbine will have an isentropic Process so it will have an imaginary stream and so we will find the properties of the imaginary Stream first.

We know that the Pressure and Temperature of stream 5 will be equal to the pressure and temperature of stream 6 because in stream 6 (the outlet of the condenser) is said to be saturated liquid at 40°C, so therefore the Pressure is the saturation Pressure at 40°C, so therefore $P_5 = P_6$ which we find using the CHE Calculator and then we can find the properties of stream 5:

- $H_5 = 40429.087 \text{ KJ/Kmol}$
- $S_5 = 129.778 \text{ KJ/Kmol} * K$
- molar enthalpy and entropy and availability and duty which we will change when we find the actual molar flow rate

Since we have an imaginary stream we can calculate the entropy balance, and we will use the entropy balance to find either x or T , and when we goal seek for the quality x to make the entropy balance equal to zero we get a quality of:

$$x_{5s} = 0.8628 \quad (39)$$

Since we have a quality we know that the temperature of stream 5 is going to be the same as stream 6 which is 40°C.

When finding the quality or the temperature for the real stream we would do the same process but we will solve for an efficiency and do an efficiency balance that we can goal seek the quality in the real stream since we know that the temperature would be equal to 40°C. So therefore when we solve for the quality we get:

$$x_5 = 0.8935 \quad (40)$$

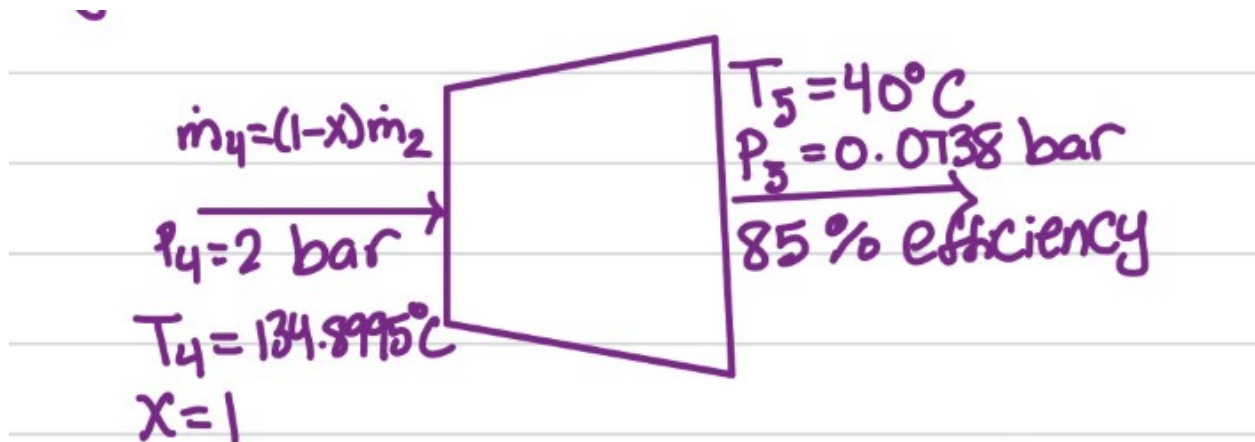


Figure 6: LP Turbine

Stream Properties	S5 Ideal	S5
Molar Flow, kmol/s	0.0252	0.0252
Mass Flow, kg/s		
Volume Flow, m ³ /s		
Temperature, C	40.0000	40.0000
Pressure, bar	0.0738	0.0738
Vapor Fraction	0.8628	0.8935
Molecular Weight, kg/kmol		
Molar Enthalpy, kJ/kmol	40429.087	41761.227
Vapor Phase	46377.193	46377.193
Liquid Phase	3016.658	3016.658
Molar Entropy, kJ/(kmol K)	129.778	134.032
Vapor Phase	148.773	148.773
Liquid Phase	10.307	10.307
Molar Density, kmol/m ³		
Enthalpy Flow, kJ/s	1018.74	1052.30
Entropy Flow, kJ/(s K)	3.27	3.38
Availability, kJ/s	37.69	39.10
Duty [Power], kJ/s [kW]	-223.76	-190.22
Efficiency		0.8500
Compressibility Factor		

Figure 7: Stream 5: Ideal and Real

4.5 The Analysis of the Condenser

The condenser is said to be saturated liquid at 40°C, to find the Pressure we will use the CHE Calculator:

$$\text{WaterPsat}(40) = 0.0738 \text{ bar} \quad (41)$$

And we find all the properties using the temperature and the phase:

$$\text{HSteam}(40, \text{L}) = 3016.658 \text{ kJ/kmol} \quad (42)$$

$$\text{SSteam}(40, \text{L}) = 10.307 \text{ kJ}/(\text{kmol} \cdot \text{K}) \quad (43)$$

$$H_6 = 3016.658 \text{ kJ/kmol} \quad (44)$$

$$S_6 = 10.307 \text{ kJ}/(\text{kmol} \cdot \text{K}) \quad (45)$$

Molar enthalpy, entropy, availability and duty will be calculated the same way as the previous devices.

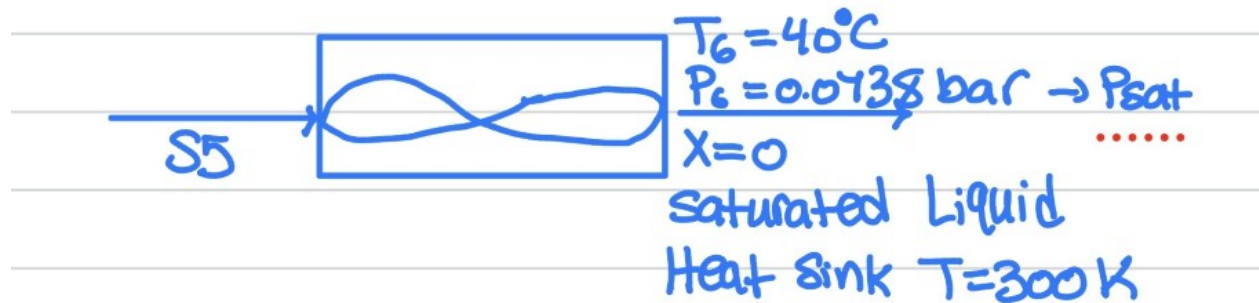


Figure 8: Condenser

13	Stream Properties	S6
14	Molar Flow, kmol/s	0.0252
15	Mass Flow, kg/s	
16	Volume Flow, m³/s	
17	Temperature, C	40.0000
18	Pressure, bar	0.0738
19	Vapor Fraction	0.0000
20	Molecular Weight, kg/kmol	
21		
22	Molar Enthalpy, kJ/kmol	3016.658
23	Vapor Phase	
24	Liquid Phase	3016.658
25	Molar Entropy, kJ/(kmol K)	10.307
26	Vapor Phase	
27	Liquid Phase	10.307
28	Molar Density, kmol/m³	
29		
30	Enthalpy Flow, kJ/s	76.01
31	Entropy Flow, kJ/(s K)	0.26
32	Availability, kJ/s	-1.90
33	Duty [Power], kJ/s [kW]	-976.29
34	Efficiency	
35		
36	Compressibility Factor	
37		

Figure 9: Stream 6

4.6 The Analysis of the Condensate Pump

The pump will also have an isentropic Process so we will have an imaginary stream so we will have the properties of it first. The condensate pump will push the Pressure out to 2 bar to make it equal to the pressure of stream 3 and that is because when entering the mixture both streams should be at equal pressures since stream 3 is at a set pressure of 2 bar we can bring the pressure of stream 7 to equal 2 bar and we also know that the product is liquid which gives us the quality of $x = 0$ and we don't know the temperatures so we would have to solve for that. We know that we can put any random numbers for T such as 40°C and find molar and flow the enthalpies and entropies as well as the availability and duty which will change when we find the actual molar flow rate.

To find the temperature of the imaginary stream we will do an entropy balance and goal seek the temperature to make the entropy balance equal to zero, so we get:

$$T_{7s} = 40.0057^\circ\text{C} \quad (46)$$

$$H_{7s} = 3020.155 \text{ kJ/kmol} \quad (47)$$

$$S_{7s} = 10.307 \text{ kJ/(kmol} \cdot \text{K)} \quad (48)$$

Now we do the actual stream with the same process but we will find an efficiency which for a pump it will be:

$$\text{efficiency} = \frac{W_{\text{ideal}}}{W_{\text{actual}}} \quad (49)$$

And we do an efficiency balance where we goal seek the temperature to set the efficiency balance equal to zero and we get:

$$T_7 = 40.0211^\circ\text{C} \quad (50)$$

$$H_7 = 3021.321 \text{ kJ/kmol} \quad (51)$$

$$S_7 = 10.311 \text{ kJ}/(\text{kmol} \cdot \text{K}) \quad (52)$$

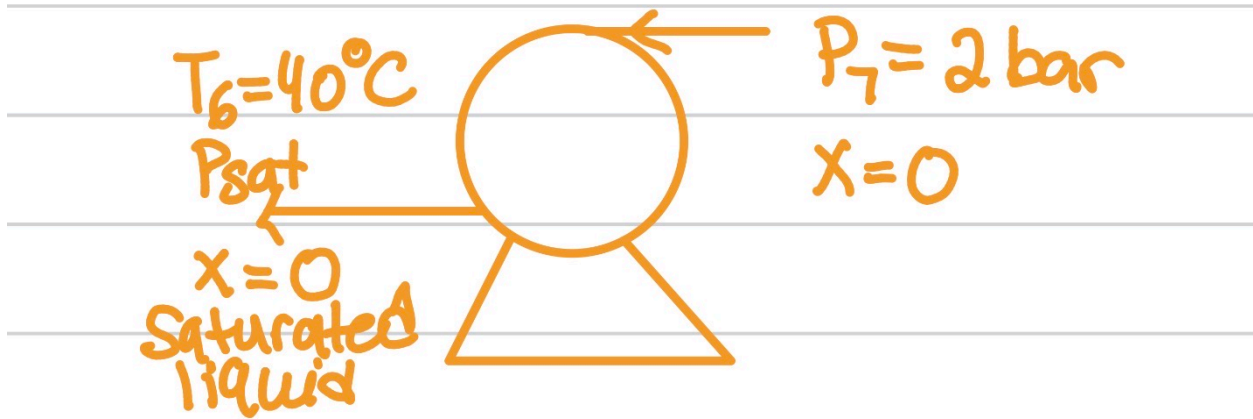


Figure 10: Condensate Pump

Stream Properties	S7 Ideal	S7
Molar Flow, kmol/s	0.0252	0.0252
Mass Flow, kg/s		
Volume Flow, m ³ /s		
Temperature, C	40.0057	40.0211
Pressure, bar	2.0000	2.0000
Vapor Fraction	0.0000	0.0000
Molecular Weight, kg/kmol		
Molar Enthalpy, kJ/kmol	3020.155	3021.321
Vapor Phase		
Liquid Phase	3020.155	3021.321
Molar Entropy, kJ/(kmol K)	10.307	10.311
Vapor Phase		
Liquid Phase	10.307	10.311
Molar Density, kmol/m ³		
Enthalpy Flow, kJ/s	76.10	76.13
Entropy Flow, kJ/(s K)	0.26	0.26
Availability, kJ/s	-1.81	-1.81
Duty [Power], kJ/s [kW]	0.09	0.12
Efficiency		0.7500
Compressibility Factor		

Figure 11: Stream 7: Ideal and Real

4.7 The Analysis of the Mixer / Open Feed water Heater

In the mixer we have two streams coming in streams 3 and 7, superheated vapor and saturated liquid. We know that the mixture doesn't change the pressure so it will stay at 2 bar. We know that it is sub-cooled liquid at 1°C so the temperature will be the saturation temperature at 2 bar minus 1°C :

$$\text{WaterTsats}(2) - 1 = 119.2308^\circ\text{C} \quad (53)$$

$$H_8 = 9015.728 \text{ kJ/kmol} \quad (54)$$

$$S_8 = 27.370 \text{ kJ}/(\text{kmol} \cdot \text{K}) \quad (55)$$

We know that the molar flow rate is:

$$\dot{n}_8 = \dot{n}_3 + \dot{n}_7 \quad (56)$$

Which we will be able to find along with the entropy and enthalpy flow as well as the availability and duty once we have the correct molar flow rate.

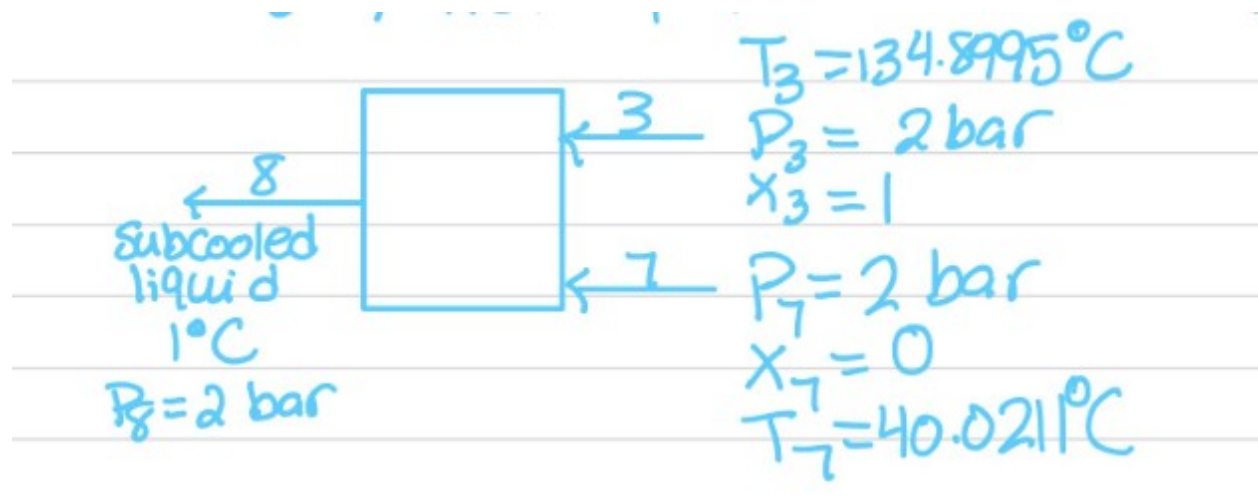


Figure 12: Mixture/Open Feedwater Heater

	Stream Properties	S8
13		
14	Molar Flow, kmol/s	0.0289
15	Mass Flow, kg/s	
16	Volume Flow, m ³ /s	
17	Temperature, C	119.2308
18	Pressure, bar	2.0000
19	Vapor Fraction	0.0000
20	Molecular Weight, kg/kmol	
21		
22	Molar Enthalpy, kJ/kmol	9015.728
23	Vapor Phase	
24	Liquid Phase	9015.728
25	Molar Entropy, kJ/(kmol K)	27.370
26	Vapor Phase	
27	Liquid Phase	27.370
28	Molar Density, kmol/m ³	
29		
30	Enthalpy Flow, kJ/s	260.98
31	Entropy Flow, kJ/(s K)	0.79
32	Availability, kJ/s	23.30
33	Duty [Power], kJ/s [kW]	0.00
34	Efficiency	
35		
36	Compressibility Factor	
37		

Figure 13: Stream 8

4.8 The Analysis of the Feed Water Pump

The pump will also have an isentropic Process so we will have an imaginary stream so we can solve for the properties of it first. we know that the outlet stream pressure is 40 bar and we also know that the product is subcooled liquid so the quality will be zero and we can even check that by doing the entropy and efficiency balances and goal seek the quality they will give us negative numbers so we know that we have to goal seek the temperature.

We know that we can plug in random numbers for T such as 120°C and find the molar and flow entropies and enthalpies and availability, and the duty, since this is an imaginary stream we don't have an efficiency. Now to find the temperature we set the entropy balance equal to zero and goal seek the temperature to make the entropy balance equal to zero so we get:

$$T_{9s} = 119.5480^{\circ}\text{C} \quad (57)$$

$$H_{9s} = 9088.222 \text{ kJ/kmol} \quad (58)$$

$$S_{9s} = 27.370 \text{ kJ/(kmol} \cdot \text{K)} \quad (59)$$

Now we do the actual stream with the same process but we will find an efficiency which for a pump it will be:

$$\text{efficiency} = \frac{W_{\text{ideal}}}{W_{\text{actual}}} \quad (60)$$

And we do an efficiency balance where we goal seek the temperature to set the efficiency balance equal to zero and we get:

$$T_9 = 119.8648^{\circ}\text{C} \quad (61)$$

$$H_9 = 9112.387 \text{ kJ/kmol} \quad (62)$$

$$S_9 = 27.431 \text{ kJ/(kmol} \cdot \text{K)} \quad (63)$$

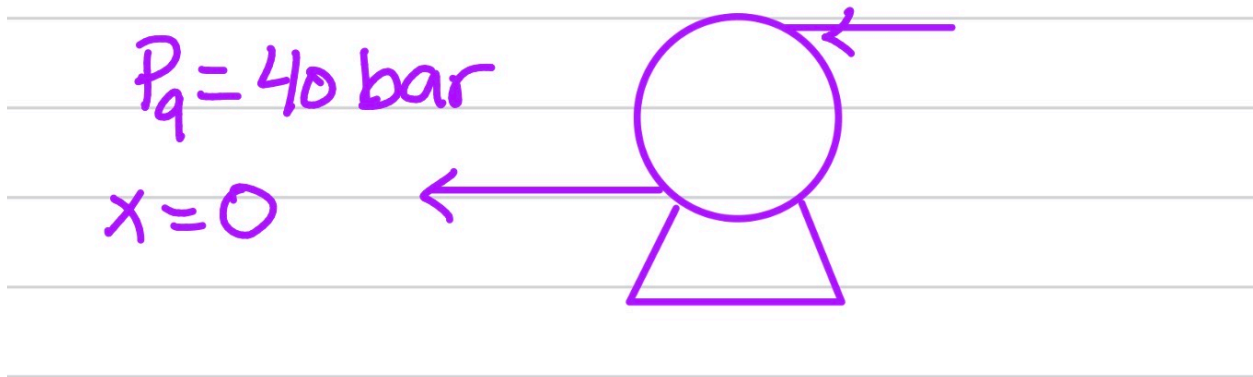


Figure 14: Feed Water Pump

Stream Properties	S9 Ideal	S9
Molar Flow, kmol/s	0.0289	0.0289
Mass Flow, kg/s		
Volume Flow, m ³ /s		
Temperature, C	119.5480	119.8648
Pressure, bar	40.0000	40.0000
Vapor Fraction	0.0000	0.0000
Molecular Weight, kg/kmol		
Molar Enthalpy, kJ/kmol	9088.222	9112.387
Vapor Phase		
Liquid Phase	9088.222	9112.387
Molar Entropy, kJ/(kmol K)	27.370	27.431
Vapor Phase		
Liquid Phase	27.370	27.431
Molar Density, kmol/m ³		
Enthalpy Flow, kJ/s	263.07	263.77
Entropy Flow, kJ/(s K)	0.79	0.79
Availability, kJ/s	25.39	25.56
Duty [Power], kJ/s [kW]	2.10	2.80
Efficiency		0.75
Compressibility Factor		

Figure 15: Stream 9: Ideal and Real

4.9 The Analysis of the Boiler (Loop)

For the boiler we have all the information needed, the temperature, Pressure, molar enthalpy and entropy, as well as the enthalpy and entropy flow but we do not have the availability or the duty but now we can calculate them since we have the information of the feed water pump, but will get the actual values when we find the power balance.

13	Stream Properties	S1
14	Molar Flow, kmol/s	0.0289
15	Mass Flow, kg/s	
16	Volume Flow, m³/s	
17	Temperature, C	450.0000
18	Pressure, bar	40.0000
19	Vapor Fraction	1.0000
20	Molecular Weight, kg/kmol	
21		
22	Molar Enthalpy, kJ/kmol	60011.902
23	Vapor Phase	60011.902
24	Liquid Phase	
25	Molar Entropy, kJ/(kmol K)	125.003
26	Vapor Phase	125.003
27	Liquid Phase	
28	Molar Density, kmol/m³	
29		
30	Enthalpy Flow, kJ/s	1737.15
31	Entropy Flow, kJ/(s K)	3.62
32	Availability, kJ/s	651.62
33	Duty [Power], kJ/s [kW]	1473.37
34	Efficiency	
35		
36	Compressibility Factor	
37		

Figure 16: Stream 1 Loop

5 Problem-Solving

5.1 Problem 1:Molar Flow Rate

Now that we have all our stream properties ready we can start solving the given problems, and we will start by calculating Problem 1 which is the molar flow rate required to provide the turbine output power of 500 kW. To solve this problem we will need a power balance:

$$- [W_{\text{turbine,LP}} + W_{\text{turbine,HP}}] - 500 \text{ kW} \quad (64)$$

We solve for that to get a number and we will use this equation to find the molar flow rate of stream one and we do that by using goal seek, where we set the power balance to equal zero by changing the molar flow rate in stream one:

$$\dot{n}_1 = 0.0289 \text{ kmol/s} \quad (65)$$

To change it to MMSCFD you use the calculator function:

$$\text{Kmol/s to MMSCFD}(0.0289) = 2.0923 \text{ MMSCFD} \quad (66)$$

5.2 Problem 2: Mass Fraction

Now we can solve Problem 2 which is the mass fraction:

$$x = \frac{H_8 - H_7}{H_3 - H_7} = \frac{9015.728 - 3021.321}{9015.728 - 49310.021} = 0.1295 \quad (67)$$

Now we can find all the other molar flow rates (in kmol/s):

$$\dot{n}_1 = 0.0289 \text{ kmol/s} \quad (68)$$

$$\dot{n}_2 = \dot{n}_1 = 0.0289 \text{ kmol/s} \quad (69)$$

$$\dot{n}_3 = x\dot{n}_2 = (0.1295)(0.0289) = 0.0037 \text{ kmol/s} \quad (70)$$

$$\dot{n}_4 = (1 - x)\dot{n}_2 = (1 - 0.1295)(0.0289) = 0.0252 \text{ kmol/s} \quad (71)$$

$$\dot{n}_5 = \dot{n}_4 = 0.0252 \text{ kmol/s} \quad (72)$$

$$\dot{n}_6 = \dot{n}_5 = \dot{n}_4 = 0.0252 \text{ kmol/s} \quad (73)$$

$$\dot{n}_7 = \dot{n}_6 = \dot{n}_5 = \dot{n}_4 = 0.0252 \text{ kmol/s} \quad (74)$$

$$\dot{n}_8 = \dot{n}_3 + \dot{n}_7 = 0.0037 + 0.0252 = 0.0289 \text{ kmol/s} \quad (75)$$

$$\dot{n}_9 = \dot{n}_8 = 0.0289 \text{ kmol/s} \quad (76)$$

Now all our enthalpy and entropy flow as well as the availability and the duty for all the streams will be solved with correct numbers.

5.3 Problem 3: Overall thermal efficiency

$$\eta_{\text{thermal}} = \frac{W_{\text{net}}}{Q_{\text{in}}} \quad (77)$$

To find Q_{in} we use the 85% boiler efficiency:

$$0.85 = \frac{Q_{\text{out}}}{Q_{\text{in}}} \quad (78)$$

$$Q_{\text{in}} = \frac{1473.51 \text{ kW}}{0.85} = 1733.823 \text{ kW} \quad (79)$$

$$\eta_{\text{thermal}} = \frac{|-309.78 + 190.22| + 0.12 + 2.80}{1733.823} \times 100\% = 29\% \quad (80)$$

5.4 Problem 4: Lost work for each device and total

We will find the lost work for each device along with the total.

Turbines and Pumps:

$$LW = B_{\text{in}} - B_{\text{out}} + \left(1 - \frac{T_0}{T}\right) \dot{Q} + W_s \quad (81)$$

There is no heat in and out so $\dot{Q} = 0$:

$$LW = B_{\text{in}} - B_{\text{out}} + W_s \quad (82)$$

Now find for each device:

HP Turbine:

$$LW = 651.62 - 300.31 + (-309.78) = 41.4689 \text{ kW} \quad (83)$$

LP Turbine:

$$LW = 261.47 - 39.10 + (-190.22) = 32.1578 \text{ kW} \quad (84)$$

Condensate Pump:

$$LW = (-1.90 - (-1.81)) - 0.12 = 0.0281 \text{ kW} \quad (85)$$

Feedwater Pump:

$$LW = 23.30 - 25.56 + 2.80 = 0.5342 \text{ kW} \quad (86)$$

Mixture and Condenser: no heat or work so

$$LW = B_{\text{in}} - B_{\text{out}} \quad (87)$$

Mixture: $B_{\text{in}} = B_3 + B_7$

$$LW = (38.90 + (-1.81)) - 23.30 = 13.7921 \text{ kW} \quad (88)$$

Condenser:

$$LW = 39.10 - (-1.90) = 40.9969 \text{ kW} \quad (89)$$

Boiler:

$$LW = B_{\text{in}} - B_{\text{out}} + \left(1 + \frac{T_0}{T_h}\right) Q_{\text{in}} \quad (90)$$

$$LW = 28.56 - 681.62 + \left(1 + \frac{300}{1250}\right) (1733.823) = 691.360 \text{ kW} \quad (91)$$

Total Lost Work:

$$41.4689 + 32.1578 + 0.0281 + 0.5342 + 13.7921 + 40.9969 + 691.360 = 820.2861 \text{ kW} \quad (92)$$

5.5 Problem 5: Thermodynamic efficiency for main goal

$$\eta_{\text{thermodynamics}} = \frac{\text{main goal}}{\text{main goal} - LW_{\text{total}}} \times 100\% \quad (93)$$

$$\eta_{\text{thermodynamics}} = \frac{-500}{-500 - 820.2861} \times 100\% = 38\% \quad (94)$$

5.6 Problem 6: Maximum theoretical efficiency

$$\eta = \left(1 - \frac{T_c}{T_h}\right) \times 100\% \quad (95)$$

$$\eta = \left(1 - \frac{300}{1250}\right) \times 100\% = 76\% \quad (96)$$

5.7 Problem 7: Annual cost of utilities

Electricity Cost, \$/year = Power of Pumps [kW] $\times$ Operating (hours/year) $\times$ Cost of electricity (\$/kWh)

$$= (0.12 + 2.80) \text{ kW} \times 8000 \text{ hr} \times 0.0775 \frac{\$}{\text{kWh}} = 1807.5938 \frac{\$}{\text{year}} \quad (97)$$

Cooling Water Cost, \$/year = Power of Condenser $\times$ hours used $\times$ Cost

$$= 76.29 \frac{\text{kJ}}{\text{s}} \times 8000 \text{ hr} \times \frac{3600 \text{ s}}{\text{hr}} \times \frac{1 \text{ kJ}}{10^3 \text{ kJ}} \times 0.3540 \frac{\$}{\text{GJ}} = 9953.4765 \frac{\$}{\text{year}} \quad (98)$$

Natural Gas Cost, \$/year = Power in Boiler $\times$ hours used $\times$ Cost

$$= 1733.3823 \frac{\text{kJ}}{\text{s}} \times 8000 \text{ hr} \times \frac{3600 \text{ s}}{\text{hr}} \times \frac{1 \text{ kJ}}{10^3 \text{ kJ}} \times 4.25 \frac{\$}{\text{GJ}} = 212165.9982 \frac{\$}{\text{year}} \quad (99)$$

Total Utilities Cost

$$= 1807.5938 + 9953.4765 + 212165.9982 = 223927.0689 \frac{\$}{\text{year}} \quad (100)$$

Problem 8: Optimization of Annual Utility Cost

For the optimization problem, I had to use a code by using the help of ChatGPT in order to design a code that can do goal seeks in the range of Pressures. The code has worked for pressures above 1 bar and therefore the code is still in the process of fixing, but since it worked for pressures above 1 bar I will use it to find the annual Cost for pressures 2–10 bar and for Pressure 1 bar I will goal seek the entire process, below are the code I used as well as a table and graph for annual Cost vs Pressure.

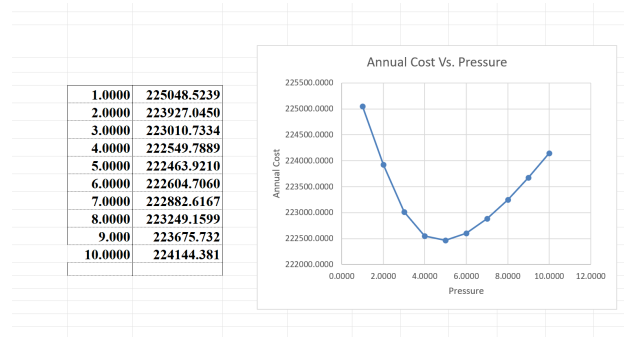


Figure 17: Annual Cost Vs. Pressure



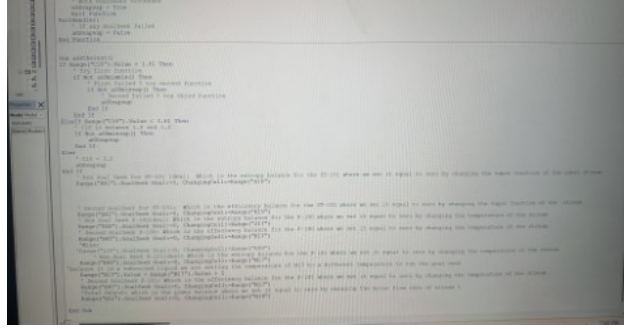


Figure 18: VBA Code

Result:

$$P_{\text{opt}} = 5 \text{ bar} \quad (101)$$

$$C_{\text{min}} = 2.22463 \times 10^5 \text{ \$/year} \quad (102)$$

Therefore, the minimum annual utility cost occurs at a high-pressure turbine discharge pressure of approximately 5 bar, corresponding to an annual utility cost of \$222,463.9210 per year.

5.8 Problem 9: The Emission of Co₂, Gt/Year



$$1733.3823 \frac{\text{kJ}}{\text{s}} \times 3600 \frac{\text{s}}{\text{hr}} \times 8000 \frac{\text{hr}}{\text{year}} \times \frac{10^3 \text{ J}}{1 \text{ kJ}} \times \frac{1 \text{ MJ}}{10^6 \text{ J}} \times \frac{1 \text{ kg}}{55.5 \text{ MJ}} = 899484.89 \text{ kg/year} \quad (104)$$

$$\begin{aligned} & 899484.89 \frac{\text{kg CH}_4}{\text{year}} \times \frac{1 \text{ kmol}}{16 \text{ kg CH}_4} \times \frac{1 \text{ kmol CO}_2}{1 \text{ kmol CH}_4} \times \frac{44 \text{ kg}}{1 \text{ kmol CO}_2} \times \frac{1 \text{ ton}}{907.1850 \text{ kg}} \times \frac{1 \text{ Gt}}{10^9 \text{ ton}} \\ & = 2.72666 \times 10^{-6} \text{ Gt/year} \end{aligned} \quad (105)$$

5.9 Problem 10: The Percentage Contribution of this Power Plant to the Worldwide Emissions.

According to the website CO₂.earth in 2020 the annual CO₂ emission was 38 Gt.

$$\frac{2.72666 \times 10^{-6} \text{ Gt/year}}{38 \text{ Gt/year}} \times 100 = 7.18 \times 10^{-6} \% \quad (106)$$

5.10 Problem 11: Natural Gas Required FLOW

Fuel used in kg/year from Problem 9:

$$899484.89 \text{ kg/year} \quad (107)$$

$$899484.89 \frac{\text{kg}}{\text{year}} \times \frac{1 \text{ year}}{365 \text{ days}} \times \frac{1 \text{ day}}{24 \text{ hr}} \times \frac{1 \text{ hr}}{3600 \text{ s}} \times \frac{1 \text{ kmol CH}_4}{16 \text{ kg CH}_4} = 1.78 \times 10^{-3} \text{ kmol CH}_4/\text{s} \quad (108)$$

Use the calculator function:

$$\text{Kmol per sec to MMSCFD}(1.78 \times 10^{-3}) = 0.1289 \text{ MMSCFD} \quad (109)$$

5.11 Problem 12: Air Requirement

Reaction:



Fuel molar flowrate:

$$1473.37 \text{ kJ/s} \left(\frac{1 \text{ MJ}}{10^3 \text{ kJ}} \right) \left(\frac{1 \text{ kg}}{55.5 \text{ MJ}} \right) \left(\frac{1 \text{ kmol}}{16 \text{ kg}} \right) = 0.0020 \text{ kmol/s} \quad (111)$$

Given:

$$y_{\text{O}_2} = 0.21, \quad y_{\text{N}_2} = 0.79 \quad (112)$$

$$x_{\text{O}_2,3} = 0.03 \quad (113)$$

Unknowns:

$$n_2 = \text{air flowrate} \quad (114)$$

$$n_3 = \text{dry flue gas} \quad (115)$$

Oxygen Balance

$$O_2 \text{ in} = 0.21n_2 \quad (116)$$

$$O_2 \text{ consumed} = 2n_1 = 2(0.0020) = 0.0039 \quad (117)$$

$$O_2 \text{ out} = 0.03n_3 \quad (118)$$

$$0.21n_2 - 0.0039 = 0.03n_3 \quad (119)$$

$$n_3 = 7n_2 - 0.13013 \quad (120)$$

Nitrogen Balance

$$0.79n_2 = x_{N_2}n_3 \quad (121)$$

Dry gas mole fractions:

$$x_{CO_2} + x_{O_2} + x_{N_2} = 1 \quad (122)$$

Since

$$n_{CO_2} = n_1 \quad (123)$$

$$x_{CO_2} = \frac{0.002}{n_3} \quad (124)$$

$$x_{N_2} = 0.97 - \frac{0.002}{n_3} \quad (125)$$

Substitute into nitrogen balance:

$$0.79n_2 = \left(0.97 - \frac{0.002}{n_3}\right) n_3 \quad (126)$$

$$0.79n_2 = 0.97n_3 - 0.002 \quad (127)$$

Insert expression for n_3 :

$$0.79n_2 = 0.97(7n_2 - 0.13013) - 0.002 \quad (128)$$

$$0.79n_2 = 6.79n_2 - 0.12623 - 0.002 \quad (129)$$

Solve for air flow:

$$6n_2 = 0.12818 \quad (130)$$

$$n_2 = 0.02136 \text{ kmol/s} \quad (131)$$

Actual Air Requirement

But our actual molar flow rate is 0.0289Kmol/s which is a little off so to find the actual air requirement in MMSCFD we will use the calculator function:

$$n_2 = 0.0289 \text{ kmol/s} \quad (132)$$

Conversion to MMSCFD:

$$\text{kmolPerSectoMMSCFD}(0.0289) = 2.09 \text{ MMSCFD} \quad (133)$$

Problem 13: The Emission of NO_x

$$\text{NO}_x = \dot{Q}_{fuel} \times \text{NO}_x \text{ factor} \quad (134)$$

Emission Factor =

$$\frac{\text{NO}_x \text{ emitted}}{\text{Fuel Energy Input}}$$

$$\text{NO}_x \text{ emitted} = \text{Emission Factor} \times \text{Fuel Energy Input} \quad (135)$$

$$= 0.050 \frac{\text{lb NO}_x}{\text{MMBtu}} \times \frac{1473.37 \text{ kW}}{0.85} \times \frac{3412 \text{ Btu/hr}}{1 \text{ kW}} \times \frac{1 \text{ MMBtu}}{10^6 \text{ Btu}} \quad (136)$$

$$= 2.96 \times 10^{-1} \frac{\text{lb NO}_x}{\text{hr}} \times 0.4536 \frac{\text{kg}}{\text{lb}} \times 8000 \frac{\text{hr}}{\text{year}} \quad (137)$$

$$= 1073.1347 \frac{\text{kg NO}_x}{\text{year}} \quad (138)$$

6 Summary of Results

6.1 Stream Property Table

Stream	$\dot{n}$	T (°C)	P (bar)	x	H (kJ/kmol)	S (kJ/kmol K)	Availability (kJ/s)	Duty (kW)
S1	0.0289	450.0000	40.0000	1.0000	60011.902	125.003	651.62	
S2 Ideal	0.0289	120.2308	2.0000	0.9664	47421.451	125.003	287.17	-364.45
S2	0.0289	134.8995	2.0000	1.0000	49310.021	129.778	300.37	-309.78
S3	0.0037	134.8995	2.0000	1.0000	49310.021	129.778	38.90	0.00
S4	0.0252	134.8995	2.0000	1.0000	49310.021	129.778	261.47	0.00
S5 Ideal	0.0252	40.0000	0.0738	0.8628	40429.087	129.778	37.69	-223.78
S5	0.0252	40.0000	0.0738	0.8935	41761.227	134.032	39.10	-190.22
S6	0.0252	40.0000	0.0738	0.0000	3016.658	10.307	-1.90	-976.29
S7 Ideal	0.0252	40.0057	2.0000	0.0000	3020.155	10.307	-1.81	0.09
S7	0.0252	40.0211	2.0000	0.0000	3021.321	10.311	-1.81	0.12
S8	0.0289	119.2308	2.0000	0.0000	9015.728	27.370	23.30	0.00
S9 Ideal	0.0289	119.5480	40.0000	0.0000	9088.222	27.370	25.39	2.10
S9	0.0289	119.8648	40.0000	0.0000	9112.387	27.431	25.56	2.80
S1	0.0289	450.0000	40.0000	1.0000	60011.902	125.003	651.62	1473.37

Table 1: Complete stream properties table .

6.2 Lost Work Summary Table

Device	Lost Work, LW (kW)
High Pressure Turbine (HP)	41.4689
Low Pressure Turbine (LP)	32.1578
Condensate Pump	0.0281
Feedwater Pump	0.5342
Mixer / Open Feedwater Heater	13.7921
Condenser	40.9969
Boiler	691.3600
Total Lost Work	820.2861

Table 2: Lost work calculated for each device and the total lost work for the cycle.

6.3 Overall Results Table (Final)

Quantity	Value
Molar Flow Rate $\dot{n}_1$	2.0923 MMSCFD
Mass Fraction x	0.1295
Thermal Efficiency	29%
Thermodynamic Efficiency	38%
Maximum Theoretical Efficiency	76%
Total Lost Work	820.2861 kW
Total Utilities Annual Cost	223927.07 \$/year
Minimum Annual Cost (5 bar)	222463.92 \$/year
Fuel Usage	0.1289 MMSCFD
CO ₂ Emissions	2.72666×10^{-6} Gt/year
Global Emission Percentage	$7.18 \times 10^{-6}\%$
Air Requirement	2.09 MMSCFD
NO _x Emissions	1073.1347 kg/year

Table 3: Overall calculated results for Problems 1–13

Stream Properties	S1	S2 Ideal	S2	S3	S4	S5 Ideal	S5	S6	S7 Ideal	S7	S8	S9 Ideal	S9	S1
Molar Flow, kmol/s	0.0289	0.0289	0.0289	0.0037	0.0252	0.0252	0.0252	0.0252	0.0252	0.0252	0.0289	0.0289	0.0289	0.0289
Mass Flow, kg/s														
Volume Flow, m ³ /s														
Temperature, °C	450.0000	120.2308	134.8995	134.8995	134.8995	40.0000	40.0000	40.0000	40.0057	40.0211	119.2308	119.5480	119.8648	450.0000
Pressure, bar	40.0000	2.0000	2.0000	2.0000	2.0000	0.0738	0.0738	0.0738	2.0000	2.0000	2.0000	40.0000	40.0000	40.0000
Vapor Fraction	1.0000	0.9664	1.0000	1.0000	1.0000	0.8628	0.8935	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
Molecular Weight, kg/kmol														
Molar Enthalpy, kJ/kmol	60011.902	47421.451	49310.021	49310.021	49310.021	40429.087	41761.227	3016.658	3020.155	3021.321	9015.728	9088.222	9112.387	60011.902
Vapor Phase	60011.902	48753.8215	49310.021	49310.021	49310.021	46377.193	46377.193							60011.902
Liquid Phase		9092.186				3016.658	3016.658	3016.658	3020.155	3021.321	9015.728	9088.222	9112.387	
Molar Entropy, kJ/(kmol K)	125.003	125.003	129.778	129.778	129.778	129.778	134.032	10.307	10.307	10.311	27.370	27.370	27.431	125.003
Vapor Phase	125.003	128.390	129.778	129.778	129.778	148.773	148.773							125.003
Liquid Phase		27.564				10.307	10.307	10.307	10.307	10.311	27.370	27.370	27.431	
Molar Density, kmol/m ³														
Enthalpy Flow, kW	1737.15	1372.70	1427.36	184.84	1242.52	1018.74	1052.30	76.01	76.10	76.13	260.98	263.07	263.77	1737.15
Entropy Flow, kW/(°K)	3.62	3.62	3.76	0.49	3.27	3.27	3.38	0.26	0.26	0.26	0.79	0.79	0.79	3.62
Availability, kW	651.62	287.17	300.37	38.90	261.47	37.69	38.10	-1.90	-1.81	-1.81	23.30	25.39	25.56	651.62
Duty [Power], kW		-364.45	-309.78	0.00	0.00	-223.78	-190.22	-976.29	0.08	0.12	0.00	2.18	2.80	1473.37
Efficiency			0.85				0.8500			0.7500			0.75	
Compressibility Factor														

38	User Data and Calculations									
39	Entropy Balance Turbine ST-100	0.0000	Power Requirement, kW	500.0000	Shaft Flow, MMSCFD	2.0023				
40	Efficiency Balance Turbine ST-100	0.0000	Turbine Efficiency	0.6500	Lost Work Turbine 100, kW	41.4889				
41	Entropy Balance Turbine ST-101	0.0000	Pump Efficiency	0.7500	Lost Work Turbine 101, kW	32.1678				
42	Efficiency Balance Turbine ST-101	0.0000	Boiler Efficiency	0.8000	Lost Work Pumps 100, kW	0.0081				
43			Cost of Electricity, \$/kWh	0.0775	Lost Work Pumps 101, kW	0.5342				
44	Entropy Balance Pump P-100	0.0000	Cost of cooling water, \$/GJ	0.3540	Lost Work Motore, kW	13.7821				
45	Efficiency Balance Pump P-100	0.0000	Cost of Natural gas, \$/GJ	4.2500	Lost Work Condenser, kW	40.9848				
46	Entropy Balance Pump P-101	0.0000	Operating hours/year	8000.0000	Lost Work Boiler, kW	691.3080				
47	Efficiency Balance Pump P-101	0.0000	HHV, MJ/kg	55.5000	Lost Work Total	826.2881				
48			Nox emission factor, lb/MMBtu	0.0500						
49	The Split Ratio/ness fraction	0.1244	Tc, K	300.0000	Reality Check	0.0000				
50			TH, K	1250.0000	Thermal Efficiency	39%				
51	Power Balance	0.0000	1 lb to Kg	0.4536	Maximum Efficiency	78%				
52			1 km to Kg	907.1850	Thermodynamic Efficiency	38%				
53			Seconds in an Hour	3600.0000	Electricity Cost, \$/year	-1807.5938	negative because paying for it			
54			Days in a Year	365.0000	Cooling Water Cost, \$/year	-2993.6780	negative because paying for it			
55			Hours in a Day	24.0000	Natural Gas Cost, \$/year	-212146.9682	negative because paying for it			
56			Seconds in a Day	86400.0000	Total Cost, \$/year	-223967.0486				
57			1 kW to Btu/hr	3412.1400						
58			molar mass of CH ₄ , Kg/Kmol	16.0000						
59			molar mass of CO ₂ , Kg/Kmol	44.0000						
60			molar mass of O ₂ , Kg/Kmol	32.0000						
61										
62										
63										
64	CO2 Emission									
65	Fuel Used, Kg/Year	89648.4191	Percentage of CO2 Emissions	38.0000	year 2020	As Requirement				
66	CO2 Produced, Gt/Year	3.226445E-04	CO2 Emissions WW, Gt	7.18E-06	Q in Boiler	1733.3620	NGx Emission		1733.3620	
67	Fuel Used, Kmol/Sec	1.78E-03	% of CO2 emissions, Gt		Molar Flow Rate of Fuel	0.0020	Fuel Energy Input, kW		2.98E+01	
68	Fuel Used, MMSCFD	0.1289			Oxygen consumed	0.0039	Nox emitted, lb/hr		1873.1347	
69					Calculations done in my notebook		Nox emitted, Kg/year			
70					air flow rate	0.0289				
71						2.0696				
72										
73										

Figure 19: Excel Sheet

7 Conclusion

In this assignment a regenerative Rankine cycle was analyzed by determining the thermodynamic properties of each stream and evaluating the performance of every device in the cycle. The CHE Calculator was used to obtain steam properties while steady state material, energy, and entropy balances were applied throughout the analysis. After determining all stream conditions, the required molar flow rate was calculated to achieve the desired turbine power output of 500 kW.

The results show that regeneration improves performance by reducing lost work in several components and increasing overall efficiency. The thermal efficiency, thermodynamic efficiency, and theoretical efficiency were evaluated and compared, showing the difference between real and ideal system performance. Additionally, fuel usage and CO₂ emissions were estimated, demonstrating how energy system design connects with environmental impact. Overall, the assignment illustrates how thermodynamic modeling, numerical solving methods, and engineering assumptions work together to evaluate a practical power cycle.

References

- Chatgpt, chatgpt.com/. Accessed 13 Feb. 2026.
- Google, Google, gemini.google.com/. Accessed 13 Feb. 2026.
- McGee, Michael. “Global CO2 Emissions.” Global CO2 Emissions, 18 Mar. 2022, www.co2.earth/global-co2-emissions.